Problem Solving with Geometric Figures

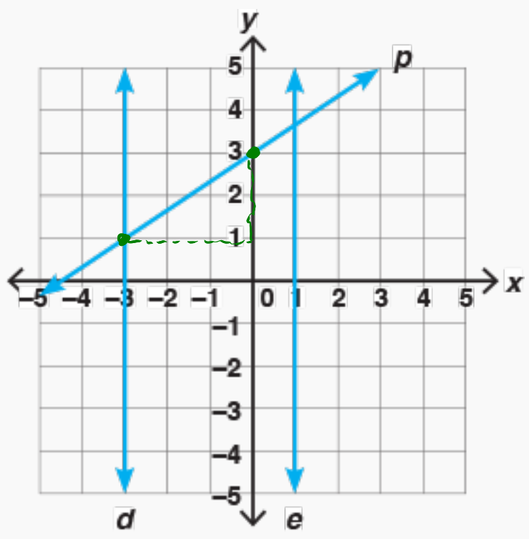
In this unit, you have gained many skills. You can plot points, draw lines, and write equations for lines drawn on a coordinate plane. You can find slope and use slope to draw parallel and perpendicular lines. You previously learned to work with similar figures, use the Pythagorean theorem to measure distance, and find perimeter and area of flat figures.

On the GED and other math tests, you will be asked to apply several skills at once to solve a problem.

Example 1

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On the coordinate plane, lines *d* and *e* are parallel. Line *p* passes through points (−3, 1) and (0, 3). Line *q* is to be drawn through point (0, −2) to complete a parallelogram. Find the equation of line *q*.



slope of $p = \frac{3-1}{0-(-3)}$

$= \frac{2}{3}$

Step 1

Apply your knowledge of polygons.

A parallelogram has two pairs of parallel lines. The new line *q* must be parallel to line *p*, so *p* and *q* must have the same slope.

Step 2

Find the slope of line *p*.

The slope of *p* is $\frac{2}{3}$ because the line rises 2 spaces as it runs 3 spaces.

Step 3

Find the equation of line *q*.

You know the slope and one point (0, −2).

Use the point-slope form: $y - y_1 = m(x - x_1)$

$y - y_1 = m(x - x_1)$

$y - (-2) = \frac{2}{3}(x - 0)$

$y + 2 = \frac{2}{3}x$

$y = \frac{2}{3}x - 2$

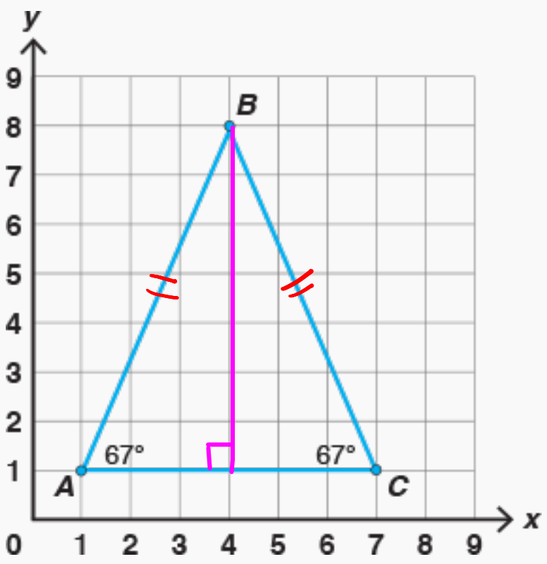
The equation of line *q* is $y = \frac{2}{3}x - 2$.

The grid marks on the coordinate plane also make it possible to solve area and perimeter problems by counting units.

Example 2

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Find the perimeter of isosceles triangle *ABC* to the nearest whole unit.



$|AB| = \sqrt{(3)^2 + (7)^2}$

$= \sqrt{9 + 49}$

$= \sqrt{58}$

≈ 7.6

Since the triangle is isosceles, *AB* is equal in measure to *BC*.

Find the distance between the points for each segment. Notice that *AB* is the hypotenuse for a triangle with a base of 3 and a height of 7.

$AB \approx 7.6 \quad BC \approx 7.6 \quad AC = 6$

Add to find the perimeter: 7.6 + 7.6 + 6 = 21.2, which rounds to **21 units.**

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Lesson 4: Problem Solving on the Coordinate Plane

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